04 · Picking funds from a retail menu: does past performance help, and what do fees cost?

Final report generated by the pipeline (python run_all.py). Tables and figures are the pipeline's own outputs; the text comes from the paper's README.

Question

An investment committee must choose which funds to offer on a menu (a company savings plan, a platform shortlist). Two questions decide the rule: do last year's winners keep winning, and do cheaper funds leave investors better off?

Result

• Last year's best fifth of equity funds beat the worst fifth by 1.11 points the following year, with a standard error of 4.61: no usable signal.

• The slope of next-year on this-year excess return is 0.094 for equity and 0.182 for fixed income, both significant but small.

• In fixed income, 39.3% of the best quintile stays best the next year, against 20% by chance; in equity, 28.0%.

• Each extra point of expense ratio goes with 0.735 points less net excess return in equity and 0.380 in multi-asset, both significant at 1%; in fixed income the slope is 0.061 and not significant.

• The cheapest third of equity funds (median expense ratio 0.16%) returned -4.99 points over the CDI the next year, against -7.18 for the most expensive third (median 2.82%).

• Over 10 years, the median equity investor who picks from the cheapest third ends with 0.72 of what the CDI would have given, against 0.69 for a random pick and 0.66 for last year's best quintile.

• Picking from the cheapest third and skipping last year's worst quintile gives the best median in every class: 0.74 in equity, 0.88 in multi-asset and 0.96 in fixed income.

Reading: past returns barely help; fees do. A menu rule of "cheapest third within each class, minus last year's worst" beats picking at random or chasing winners, but in 2021–2025 even that trailed the CDI. Four year pairs is a short history: these are associations in one period, not a law.

Data checks

7 pass, 5 warnings, 0 fail (details in output/data_checks.json).

Main tables

Next-year excess return over the CDI (p.p.) by quintile of this year's excess return (average of the four year pairs)

	Equity	Multi-asset	Fixed income
Q1 (worst this year)	-3.34	-0.67	-1.85
Q2	-2.24	-2.09	-1.36
Q3	-3.01	-2.23	-0.55
Q4	-1.07	-1.28	-0.24
Q5 (best this year)	-2.23	-1.08	-1.71
Q5 minus Q1	1.11	-0.42	0.14
	(4.61)	(2.68)	(2.36)
Slope: next on this year	0.094**	0.094**	0.182***
	(0.040)	(0.045)	(0.065)
Fund-years	3677	5314	4619

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Share of funds that stay in the same quintile next year (chance = 20%)

	Equity	Multi-asset	Fixed income
Best quintile stays best (%)	28.0	29.2	39.3
Worst quintile stays worst (%)	31.6	28.0	33.3
Worst quintile closed next year (%)	1.1	1.8	0.9

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Fees and next-year net excess return over the CDI (p.p.)

	Equity	Multi-asset	Fixed income
Expense ratio (slope)	-0.735***	-0.380***	0.061
	(0.189)	(0.127)	(0.080)
Performance fee (dummy)	-5.120	1.550	-0.050
	(3.528)	(3.273)	(0.653)
Observations	3432	4774	4411
R^2	0.365	0.019	0.066
Cheapest third: next-year excess	-4.99	-1.72	-1.30
Cheapest third: median expense ratio	0.16	0.07	0.04
Middle third: next-year excess	-4.15	-1.48	-1.07
Middle third: median expense ratio	1.22	1.10	0.36
Most expensive third: next-year excess	-7.18	-2.59	-1.53
Most expensive third: median expense ratio	2.82	2.95	1.17

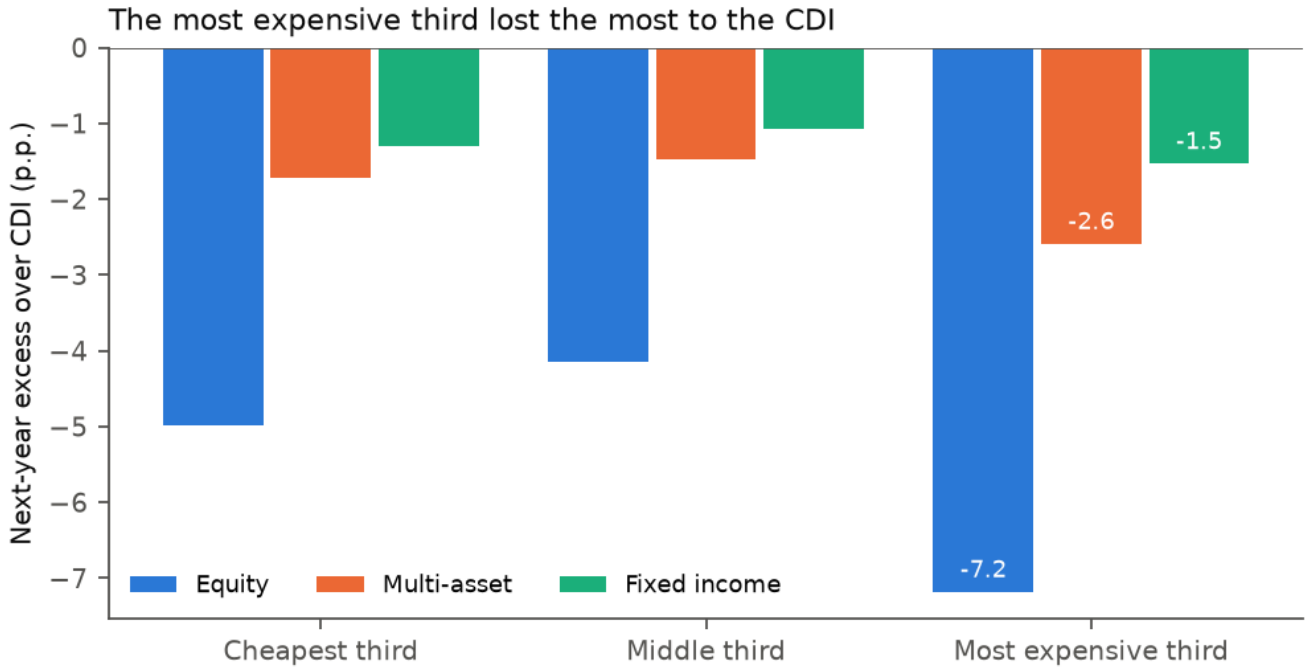
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Monte Carlo over 10 years: wealth relative to staying in the CDI (1.00 = same as the CDI), 10000 paths

Class: rule	Median	P5	P95	Beats CDI (%)
Equity: Random fund	0.69	0.23	1.89	27
Equity: Cheapest third	0.72	0.24	2.01	30
Equity: Last year's best quintile	0.66	0.20	2.07	29
Equity: Cheapest third, not in worst quintile	0.74	0.26	1.94	31
Multi-asset: Random fund	0.85	0.47	1.44	28
Multi-asset: Cheapest third	0.87	0.40	1.50	32
Multi-asset: Last year's best quintile	0.82	0.35	1.81	32
Multi-asset: Cheapest third, not in worst quintile	0.88	0.54	1.42	30
Fixed income: Random fund	0.92	0.72	1.05	14
Fixed income: Cheapest third	0.93	0.64	1.07	22
Fixed income: Last year's best quintile	0.86	0.62	1.07	15
Fixed income: Cheapest third, not in worst quintile	0.96	0.75	1.05	27

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Main figures



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Monte Carlo: median (dot) and 5th-95th percentile (line); 1.00 = same as the CDI



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